

Auto-Ranging Capacitance Meter on the ATmega328P

Build, setup, results, and conclusions

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A digital capacitance meter built on an ATmega328P. An unknown capacitor is charged through a known resistor; the time to cross one-third and two-thirds of the supply is timed in hardware, making the reading independent of supply voltage ($C = \Delta t / (R \ln 2)$). A 16-channel analog multiplexer selects one of eleven resistors for automatic ranging over seven decades, and a two-point calibration corrects the residual error. Capacitors from 100 pF to 470 μ F were measured; the dominant low-range limit — multiplexer on-resistance ($\sim 150 \Omega$) — is identified and quantified.

1 Objective

To build an instrument that measures capacitance with an 8-bit microcontroller, which cannot read capacitance directly but can time intervals precisely. The unknown capacitor is converted into a charge-time, measured by a hardware timer, displayed locally, and streamed over Bluetooth — with **automatic ranging** so a single device covers many decades of capacitance.

2 Method and setup

2.1 Measurement principle

A capacitor C_x is charged through a known resistor R ; its voltage follows $v(t) = V_{cc}(1 - e^{-t/RC})$. Instead of timing to a single level (which would depend on V_{cc}), we time the **interval between two supply-referenced thresholds**, $\frac{1}{3}V_{cc}$ and $\frac{2}{3}V_{cc}$. The supply then cancels:

$$\Delta t = RC \ln \frac{V_{cc} - \frac{1}{3}V_{cc}}{V_{cc} - \frac{2}{3}V_{cc}} = RC \ln 2 \quad \Rightarrow \quad \boxed{C = \frac{\Delta t}{R \ln 2} = \frac{\Delta t}{0.693 R}}$$

So C depends only on a known resistor, a measured time, and a constant — not on the supply, the comparator's trip point, or temperature.

2.2 What was built

```
Cx  R(range)  node  analog comparator  Timer1 input capture
      CD4067 mux      &  Vcc divider
      (auto-range R)
```

- **MCU:** ATmega328P at 8 MHz. The on-chip **analog comparator** drives the **Timer1 input-capture** unit, which latches the charge time in hardware (no interrupt jitter).
- **Thresholds:** the comparator's negative input is switched via the ADC multiplexer between $\frac{1}{3}V_{cc}$ and $\frac{2}{3}V_{cc}$ taps from a three-resistor divider; only one register write changes the threshold.
- **Auto-ranging:** a **CD74HC4067** 16-channel mux selects one of **eleven measured resistors** (**10 Ω – 5.6 M Ω**). Firmware picks the channel that keeps the timer count in a good window ($200 \leq N \leq 58000$ ticks), using a warm-start + interpolation search that converges in $\sim 1\text{--}3$ trials.
- **Output:** a 4-digit seven-segment display driven over hardware SPI through shift registers, plus an **HC-06 Bluetooth** link that streams readings and lets a host PC trigger measurements.

Signal	Pin	Role
Charge / discharge	PB0	HIGH = charge through R , LOW = discharge
Node sense	PD6 (AIN0)	comparator (+)
Thresholds	PC4 / PC3	$\frac{1}{3}V_{cc}$ / $\frac{2}{3}V_{cc}$ (comparator –, via ADC mux)
Range select	PC0–PC2, PD7	CD4067 channel (S0–S3)
Display / BT	PB2,3,5 / PD0,1	SPI to 595s / HC-06 UART

2.3 Procedure

Each reading discharges the capacitor, then times one charge from $\frac{1}{3}$ to $\frac{2}{3}V_{cc}$; **30 cycles are taken and a trimmed mean** (drop 3 high + 3 low) reported, suppressing comparator jitter. The tick count is converted with a calibrated **affine** law $C = a \Delta t + b$, where the slope absorbs the clock and resistor errors and the intercept absorbs stray capacitance. The constants a, b are fixed by **two known capacitors** (a single point cannot, as a non-zero offset is present).

3 Results

3.1 Calibration (single range, $R = 99.3 \text{ k}\Omega$)

Trimmed-mean intervals for known capacitors gave a clean straight line with a **non-zero intercept**, confirming the affine model. The two-point fit (10 nF, 100 nF):

$$a = \frac{100000 - 10000}{7788 - 733} = 12.76 \text{ pF/tick}, \quad b = +650 \text{ pF}.$$

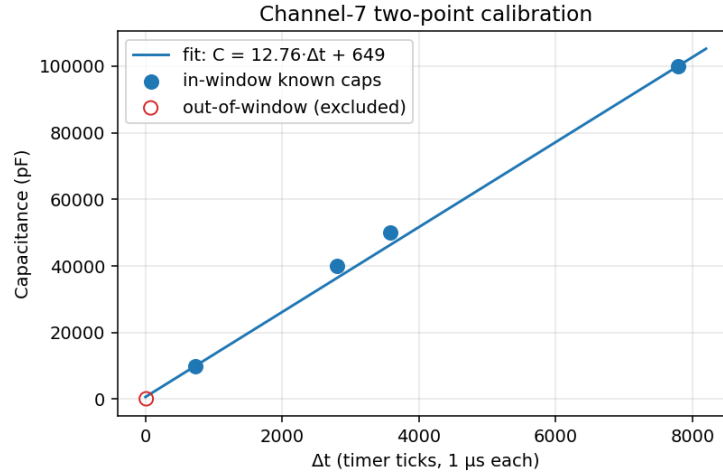


Figure 1: Channel calibration: in-window points (filled) define the affine fit; the +650 pF intercept is the stray offset. The 100 pF point (open) is out of window and excluded.

Each point is the **trimmed mean of 30 charge cycles** (3 highest + 3 lowest dropped); the median is shown as a cross-check of repeatability:

Capacitor	mean Δt	min	max	median	spread
100 pF	12	12	13	12	(out of window)
10 nF	733	696	760	736	8.7 %
104 ceramic	2802	2665	3360	2802	24.8 %
50 nF	3579	2943	4048	3587	30.9 %
100 nF	7788	6780	9228	7777	31.4 %

The complete 30-cycle runs (Δt in ticks; comparator jitter visible, hence the trimmed mean; **bold** = the two calibration anchors):

```

100pF: 13 12 12 12 12 12 13 13 13 13 12 12 12 12 12
      12 12 12 12 12 12 12 12 13 12 13 13 13 12 12
10nF : 732 699 709 735 746 749 744 731 759 738 721 707 719 729 732
      737 760 749 736 734 742 727 696 715 740 740 738 713 755 744
104  : 3360 2700 2881 2809 2826 2744 2774 2903 2799 2811 2828 2802 2734 2813 2882
      2777 3082 2790 2839 2755 2795 2698 2790 2665 2771 3088 2670 2851 2812 2774
50nF : 3676 3637 3590 3695 3651 3516 3661 3706 3344 4048 3572 3587 3550 3491 3501
      3495 3423 3508 3547 3510 3721 3511 3714 3601 3648 2943 3493 3360 3616 3758
100nF: 9228 7637 7661 7940 8097 7713 7864 7524 7536 8687 7775 7954 8048 7912 7807
      7972 7681 7687 7586 7674 7837 8230 7877 7866 7777 7759 6873 6780 7659 7594

```

->12
->733 *
->2802
->3579
->7788 *

3.2 Auto-ranging (100 pF – 470 μF)

The meter selected an appropriate range for every capacitor across seven decades:

Capacitor	Channel	Reading	Note
0.33 μF	R3 (980 Ω)	0.337 μF	accurate
1 μF	R5 (9.86 kΩ)	0.975 μF	accurate

Capacitor	Channel	Reading	Note
10 μF	R4 (5.5 k Ω)	8.76 μF	within electrolytic tolerance
47 μF	R3	46.5 μF	accurate
220 μF	R1 (56 Ω)	715 μF	error — see below
680 μF	R0 (10 Ω)	overflow	beyond range

The range-selection logic was correct in every case; accuracy is excellent wherever the multiplexer resistance is negligible.

3.3 Key finding: multiplexer on-resistance

The errors at the smallest resistors are caused by the multiplexer’s series **on-resistance** R_{on} . From the 220 μF reading, $R_{\text{eff}} = \Delta t / (C \ln 2) \approx 207 \Omega$ versus a nominal 56 Ω , so $R_{\text{on}} \approx 150 \Omega$ — matching the datasheet. The error scales as $(R_{\text{nom}} + R_{\text{on}}) / R_{\text{nom}}$: negligible (+0.15 %) at 99 k Ω but $\sim 3.7\times$ at 56 Ω , which explains both the 220 μF over-read and the 680 μF overflow. A software calibration model independently confirmed $R_{\text{on}} \approx 75\text{--}150 \Omega$ and a $\sim 600 \text{ pF}$ stray offset.

4 Conclusions

A supply-independent, auto-ranging capacitance meter was successfully built and validated from **100 pF to 470 μF** . The two-threshold method cancels the supply voltage, automatic ranging keeps every measurement at good timer resolution, and a two-point affine calibration corrects the systematic error in the mid range.

The dominant limitation is the multiplexer **on-resistance** ($\sim 150 \Omega$), which degrades the low-resistance (large-capacitor) ranges. Three improvements follow directly: (i) use **measured effective resistances** ($R_{\text{nom}} + R_{\text{on}}$) in the calibration table; (ii) switch to a **coarser timer prescaler** on the large-capacitor ranges to avoid overflow; and (iii) calibrate with **film reference capacitors across ranges** (and a crystal timebase) for uniform accuracy across all seven decades.